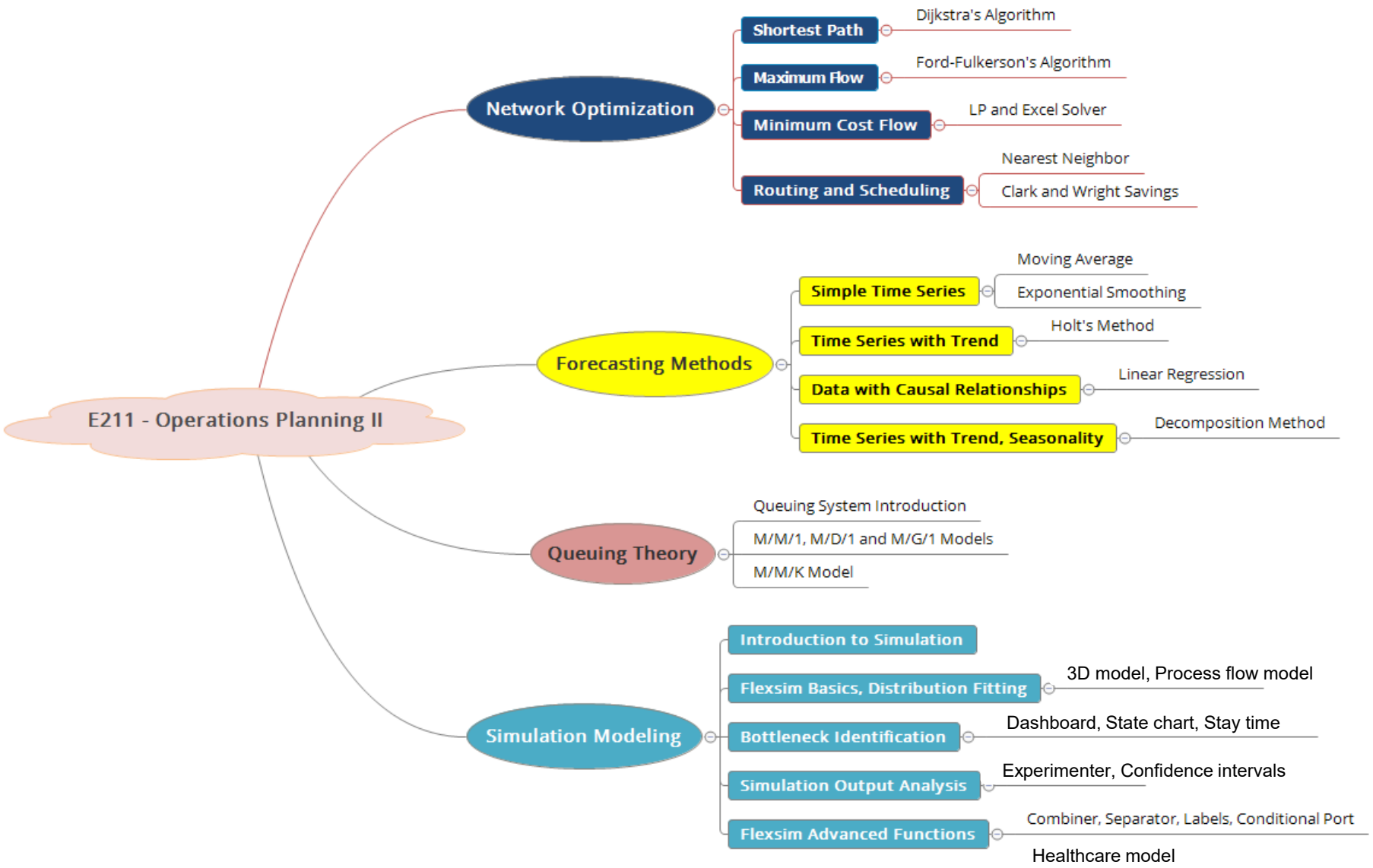
REPUBLIC POLYTECHNIC

# Lesson 7
# Modeling & Simulation of Bottling Line (I)

**E211 – Operations Planning II**

REPUBLIC POLYTECHNIC

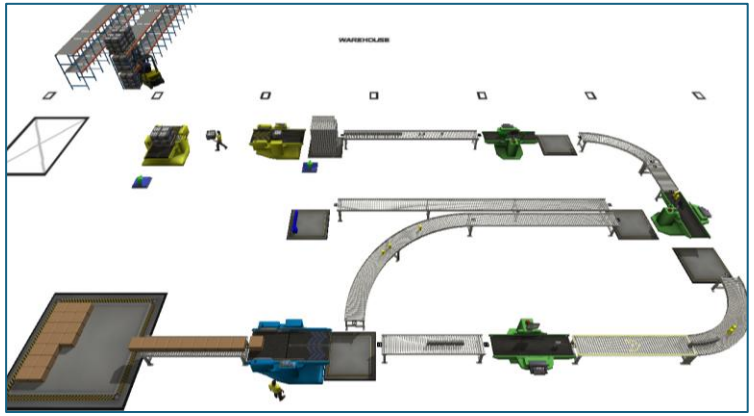
# E211 Topic Tree



# Learning Outcomes

At the end of the lesson, students should be able to:

- Construct a 3D production model in FlexSim to simulate material conveyance and processing.
- Apply modeling features in FlexSim to simulate differentiated flow, status tracking and appearance alteration.
- Perform multiple simulation runs to generate mean and confidence interval results.



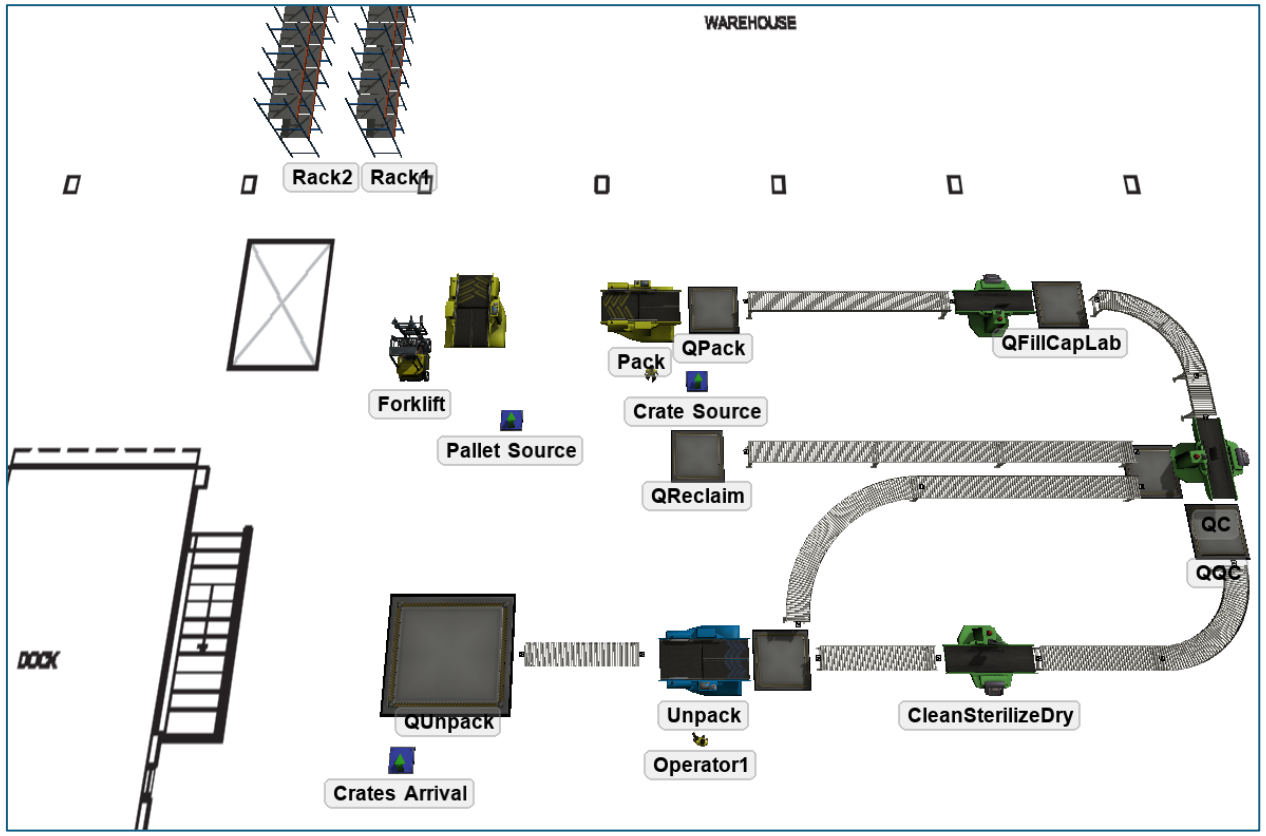
# Case Scenario

JuiceBoost, a manufacturer of organic vegetable and fruit juice blends, supplies its products to restaurants and supermarkets. Sales have been increasing substantially, and the company plans to build another bottling line for a new brand of super juice.

Justin, the production supervisor, is tasked to evaluate the new bottling line. By working with a simulation team, he wants to study options to improve the proposed bottling line, including ways to reduce the process time and the bottle reclaim rate.

*Refer to 'AY 2025 Sem 2 E211 L7 Scenario' for detailed description of the proposed bottling line.*

# Completed Model in FlexSim



# Production Line Performance Measures

For the bottling production line, the following performance measures will be measured:

- Completion time for a given production order
- Number of cartons produced
- Number of reclaims

# Required Data to Build the Model

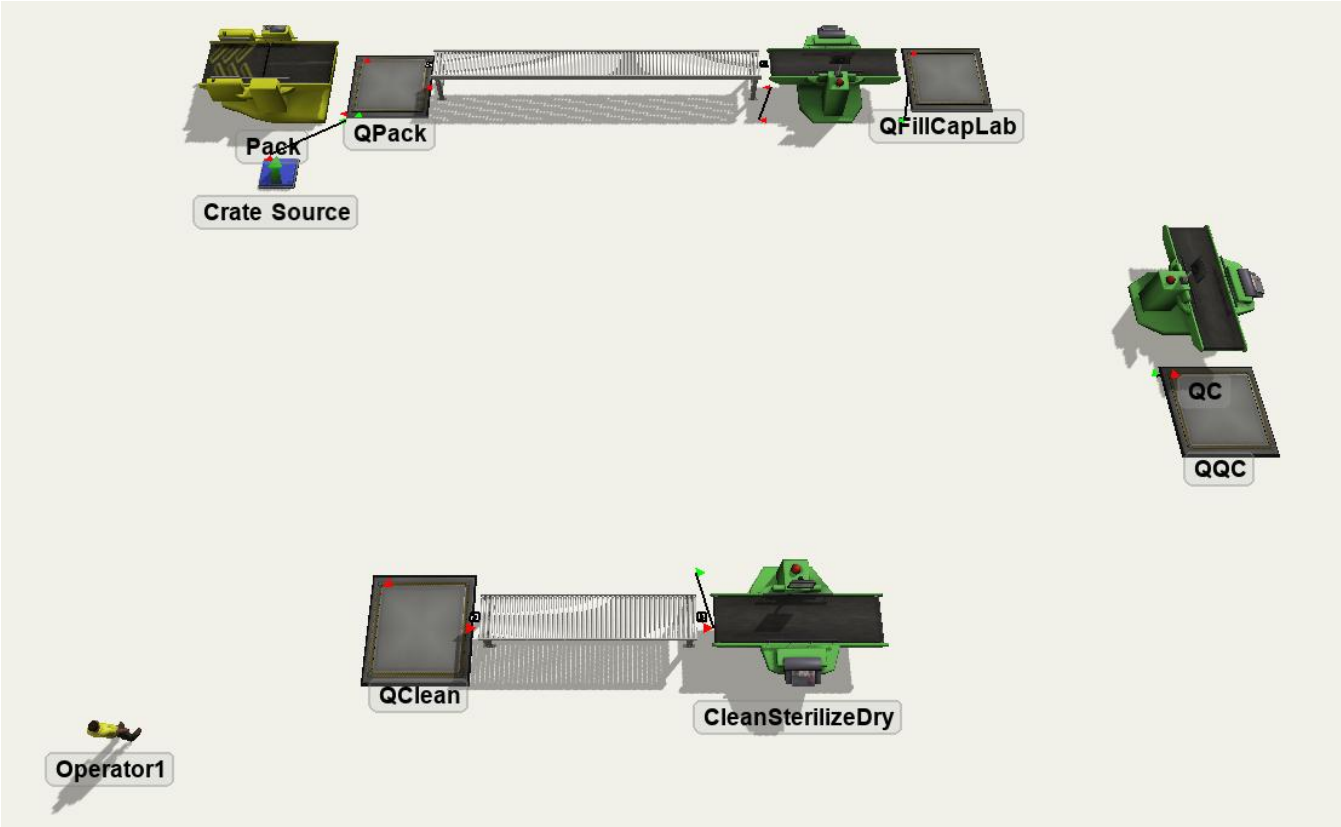
- Facility layout and dimensions
- Product info (dimensions, UOM)
- Production order (Incoming number of bottles)
- Setup and process time of each production step
- Percentage of bottles that fail quality check
- Characteristics (such as speed & capacity) of operators and material handling equipment - forklift and conveyors

# Model Building Steps

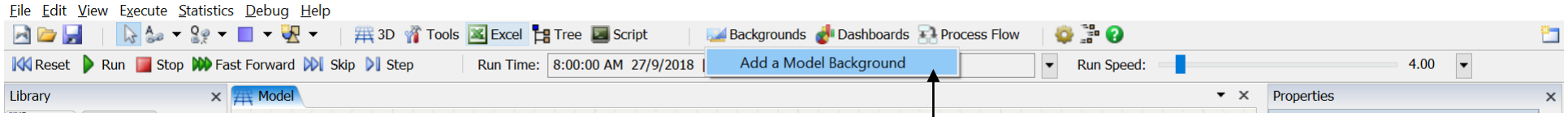
1. Import model background
2. Create item arrivals
3. Use Separator to model unpacking
4. Connect production flow with conveyors
5. Set Processor parameters
6. Create and use item QC label
7. Set item color with QC label
8. Use Combiner object to model packing
9. Use Combiner object to model palletization
10. Create storage Racks
11. Model Task Executers
12. Create Dashboard
13. Set up and run Experimenter

# Open the Starting Model

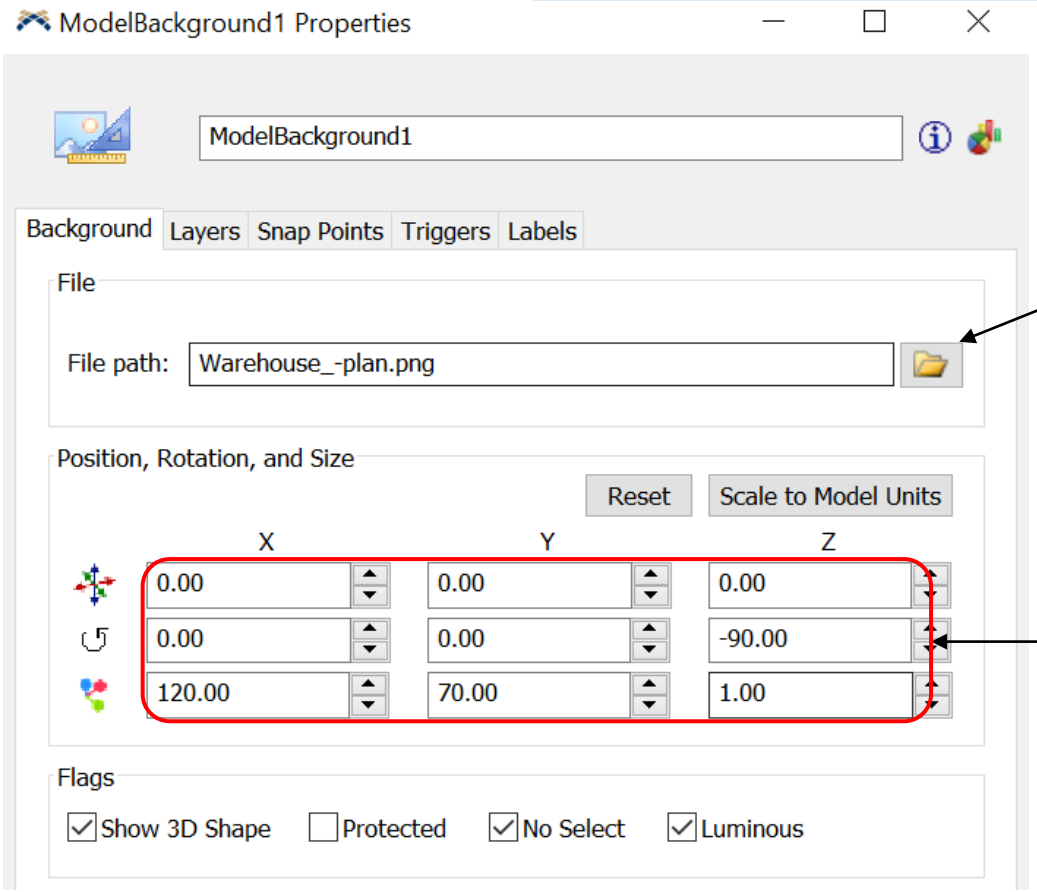
Open the FlexSim file ‘AY 2025 E211 L7 Starting Model’.



# 1. Import Model Background




1. Click Backgrounds and Add a Model Background

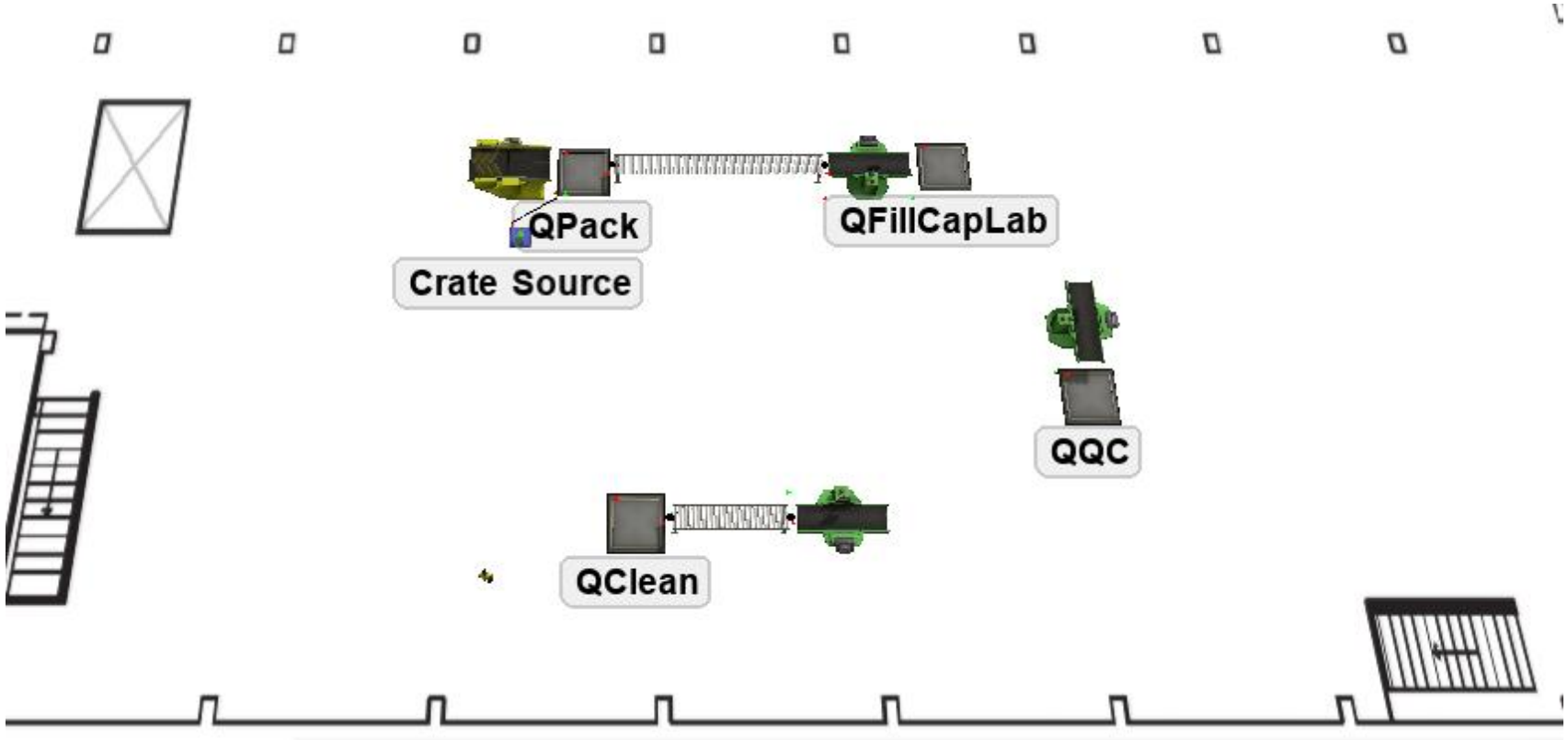


Warehouse_-plan.png

2. Save file to PC first before selecting file path

3. Enter the values as shown

# Model Background Imported




You should see the background aligned this way.

# 2. Create Item Arrivals

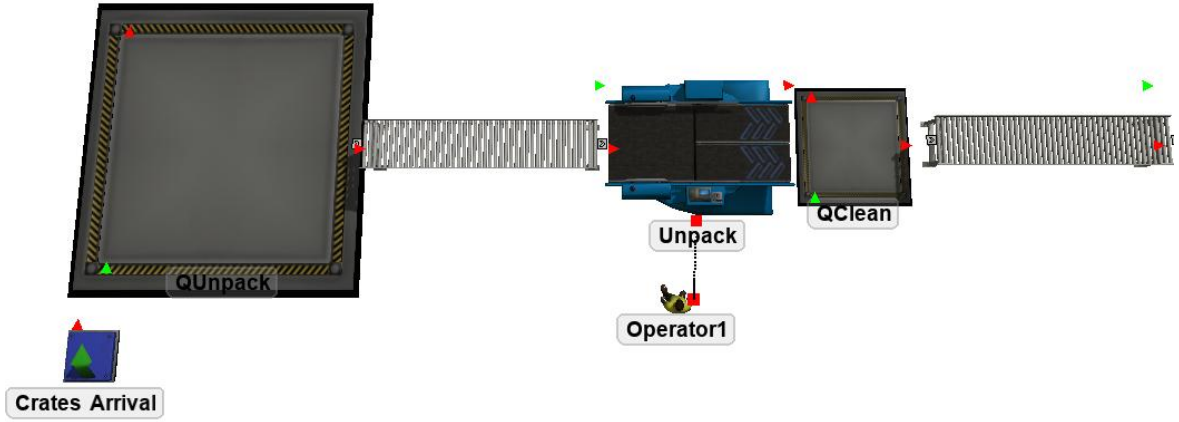
Create the following objects and position them as shown:

a. **Source** (Rename as ‘Crates Arrival’)

b. **Queue** (Size: 5 x 5. Rename as ‘QUnpack’)

c. **Separator** (Rename as ‘Unpack’)

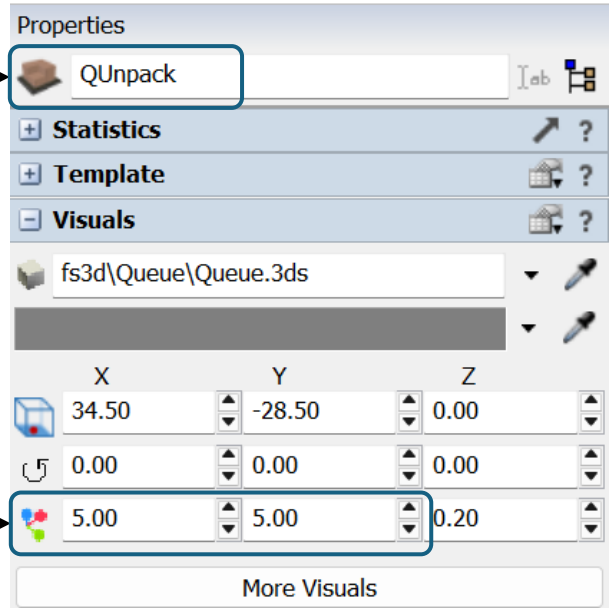
d. **Conveyor** (Length ~ 5m between QUnpack and Unpack)




Properties page of Queue

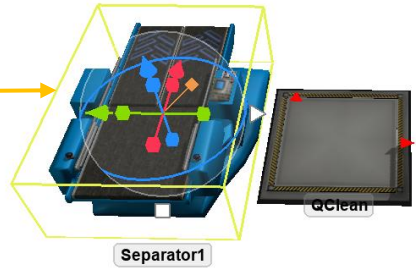
Rename here

Set the size of Queue




**To rotate an object, click to select it.**

**Then click the circle around it and rotate.**

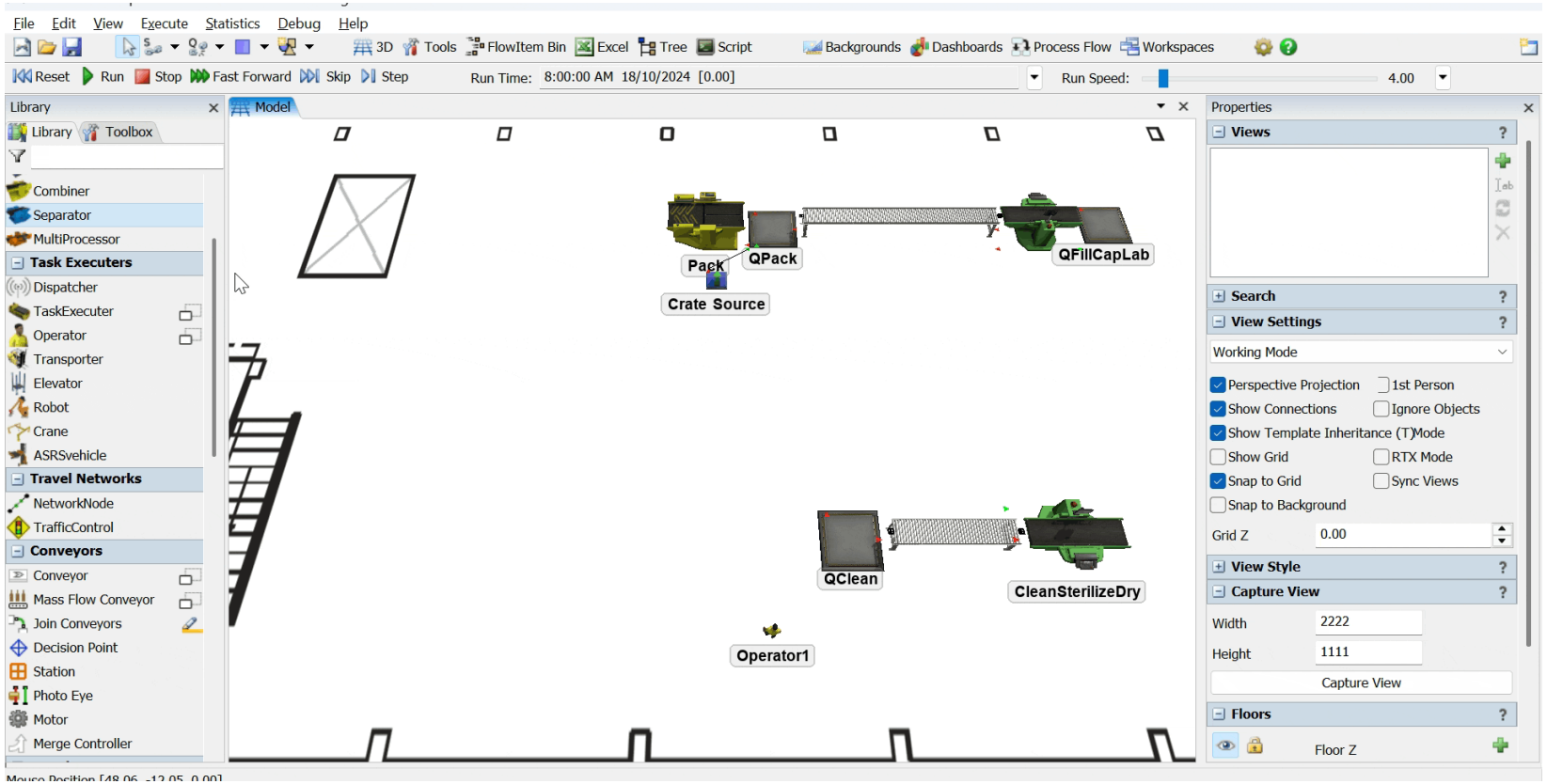


# Connect Objects

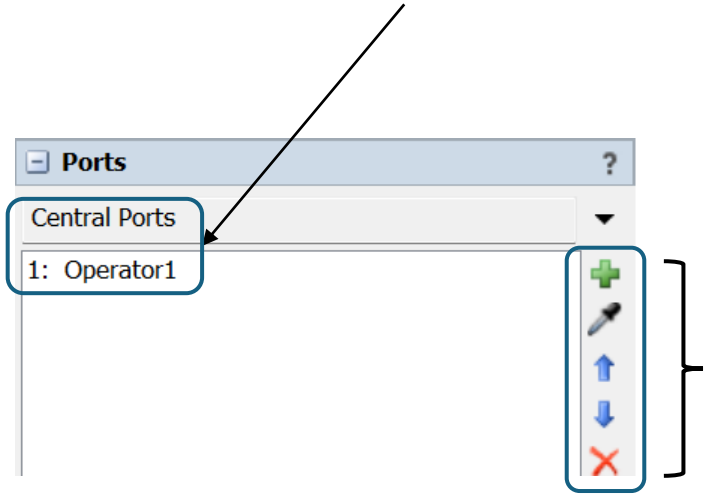
- Connect the objects using ‘Connect Objects (A)’

Crates Arrival → QUnpack → Conveyor → Unpack → Qclean

- Connect Operator1 to Unpack using ‘Connect Center Ports (S)’




Under Ports section in properties, check that Operator1 is connected to Unpack via Central Port.




Icons to connect, disconnect and change port number

# Production Schedule

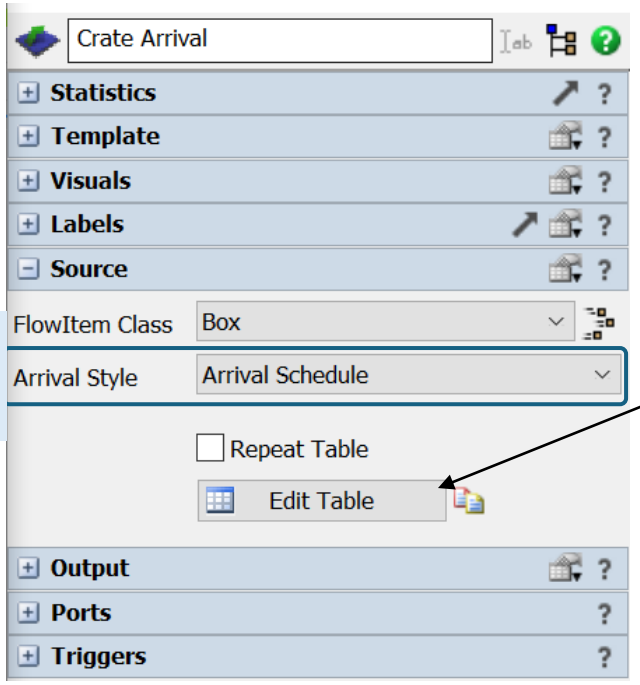
- In the production line, the arrivals occur at a fixed schedule.
- A batch of 60 crates of bottles will arrive at the production line every half an hour.
- A total of 13 batches is scheduled on a normal day.
- Arrivals of bottles can be generated from the **Source** object.

# Model Item Arrivals in Source

**Arrival Style -**

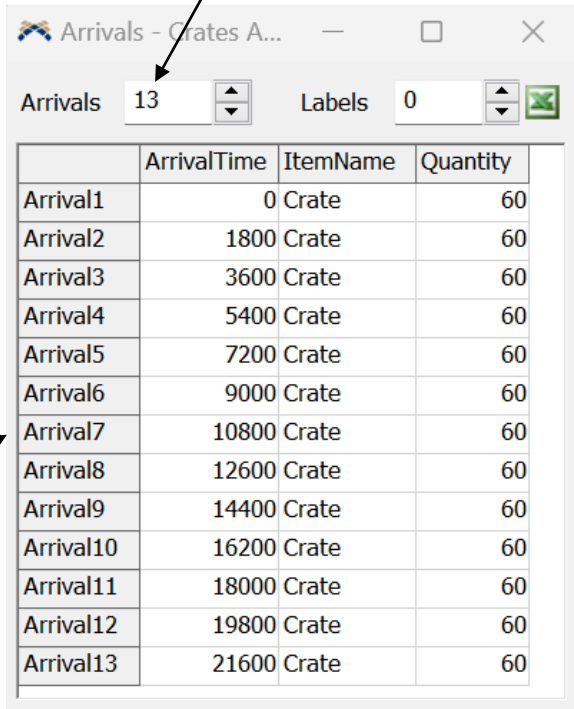
- Inter-Arrival Time: Time between successive arrivals
- Arrival Schedule: Exact arrival times and quantities
- Arrival Sequence: Exact quantities in order of arrivals

1. Change Arrival Style to **Arrival Schedule**




2. Click Edit Table to bring up the Arrivals table

3. Enter '**13**' in Arrivals

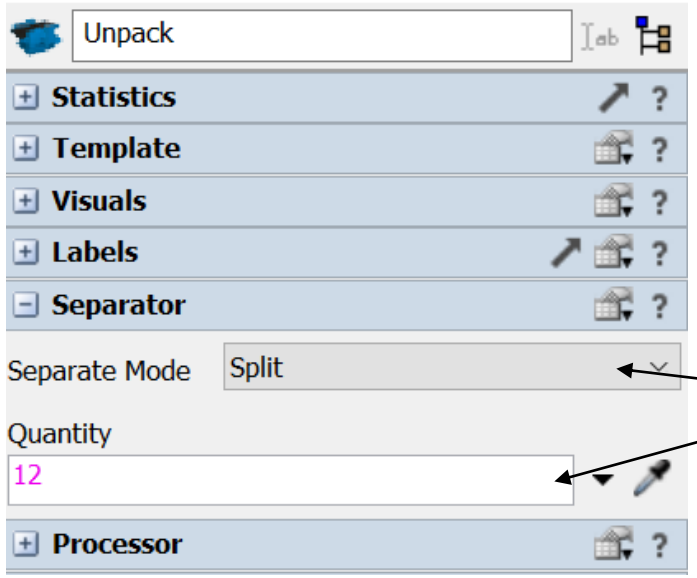



| | ArrivalTime | ItemName | Quantity |
|---|---|---|---|
| Arrival1 | 0 | Crate | 60 |
| Arrival2 | 1800 | Crate | 60 |
| Arrival3 | 3600 | Crate | 60 |
| Arrival4 | 5400 | Crate | 60 |
| Arrival5 | 7200 | Crate | 60 |
| Arrival6 | 9000 | Crate | 60 |
| Arrival7 | 10800 | Crate | 60 |
| Arrival8 | 12600 | Crate | 60 |
| Arrival9 | 14400 | Crate | 60 |
| Arrival10 | 16200 | Crate | 60 |
| Arrival11 | 18000 | Crate | 60 |
| Arrival12 | 19800 | Crate | 60 |
| Arrival13 | 21600 | Crate | 60 |

4. Key in the data as shown. The data can be prepared in Excel first and then copied here.

# 3. Model Separator for Unpacking

To simulate unpacking of bottles from a crate, a Separator object is used.

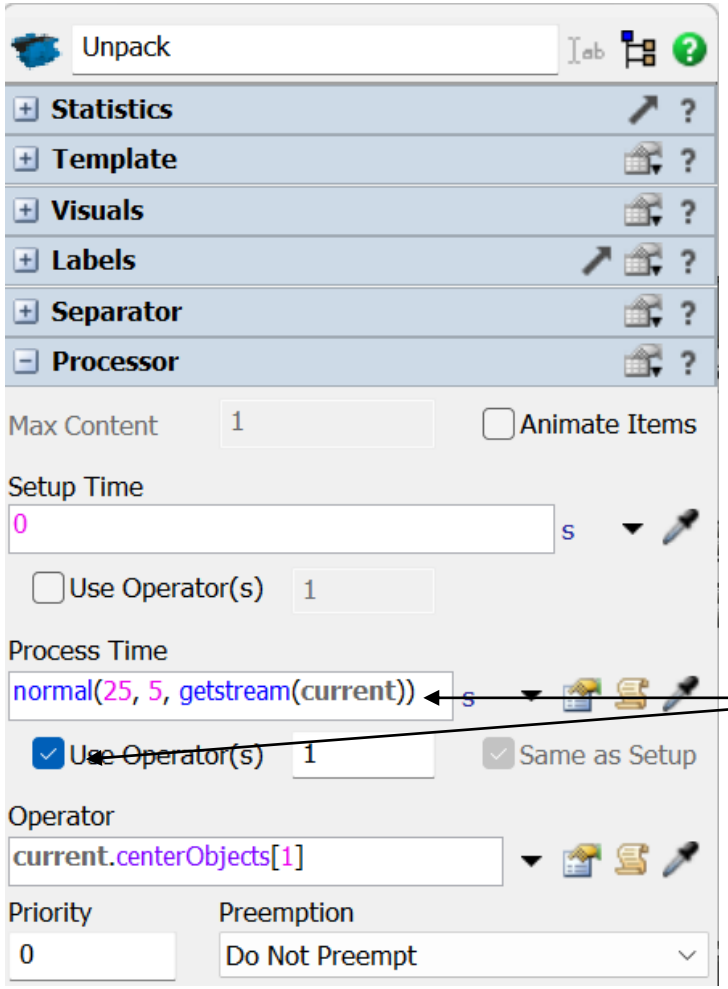



**Separate Modes**:

- **Unpack**: Unpack a container (such as totes, pallets) with packed items
- **Split**: Make multiple copies of the original item

1. Choose **Split** option
2. Enter '12' in Quantity

# Set Separator Process Time




1. At Process Time, select Statistical Distribution → **Normal distribution**. Key in mean = 25 & std dev = 5.
2. Check Use Operator.

# Alter Item ('Bottle') Shape, Color & Size

Change 3D Shape

Object: item

Shape: 2 - fs3d/General/Cylinder.3ds

Note: The getshapeindex command, used in this trigger, is slow if called frequently

Go to Triggers, create an On Exit trigger.
Under Visual, add:

1. Change 3D Shape – Select shape '**2-Cylinder**' for bottle image.

Set Object color

Object: item

Color: Color.gray

2. Set Object Color – Select **Color.gray**

Set Location, Rotation, or Size

Set: Size

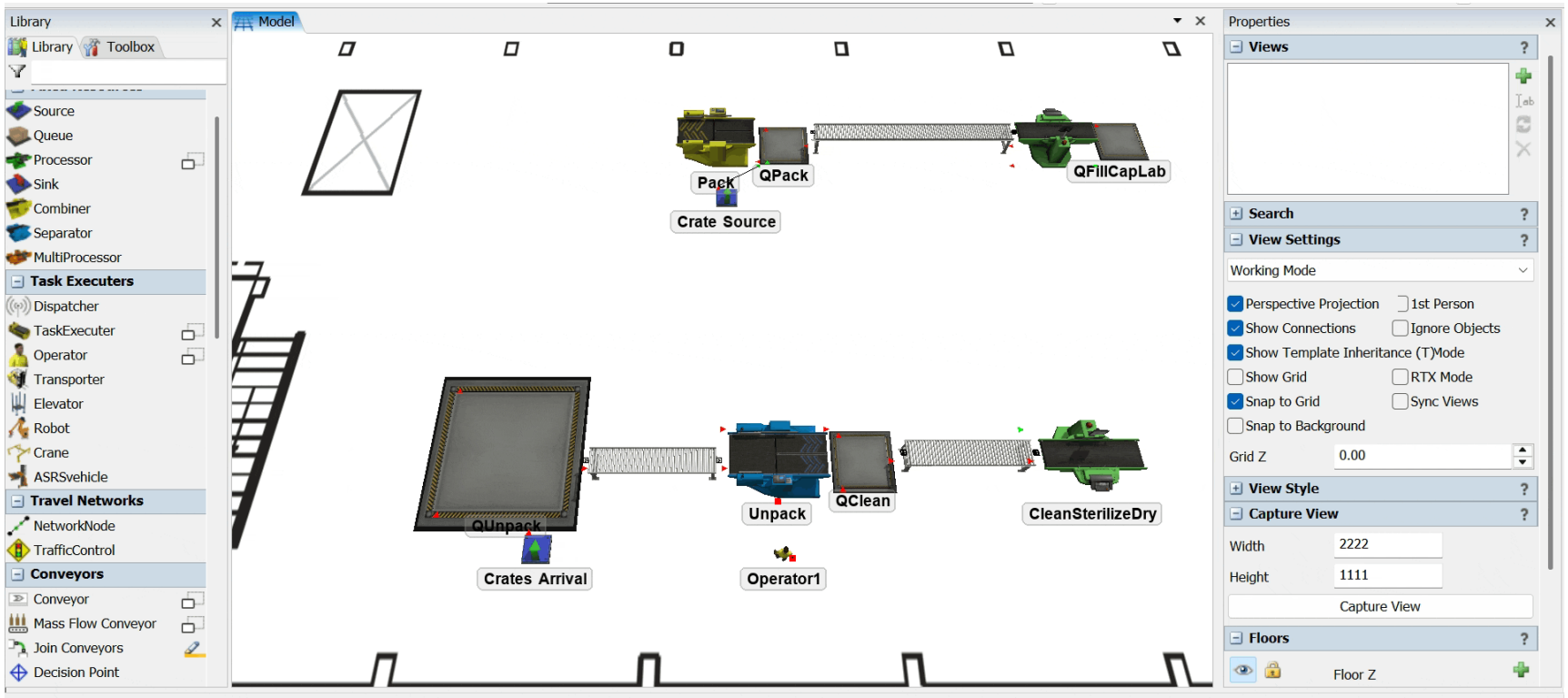
Object: item

X Size: 0.15

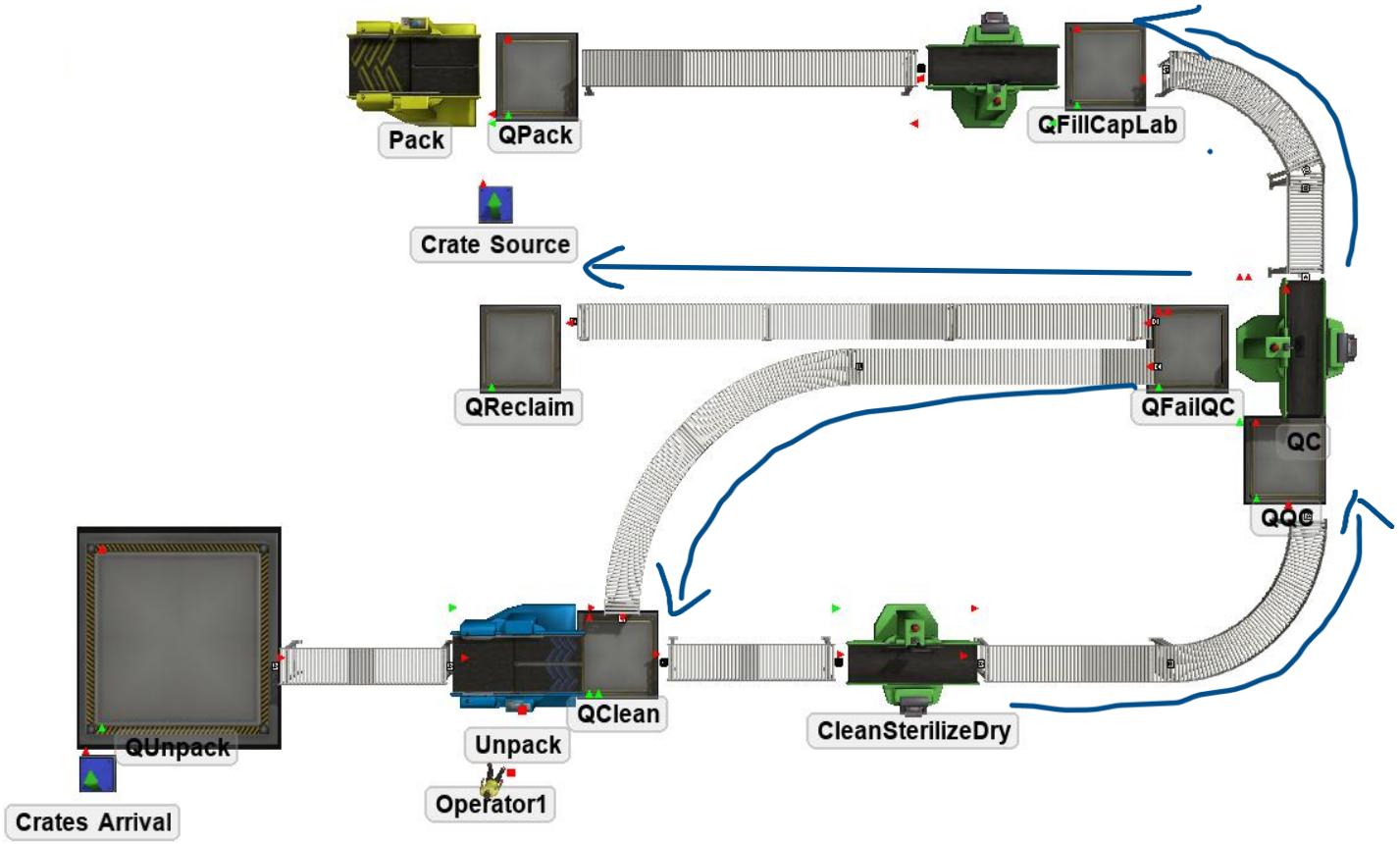
Y Size: 0.15

Z Size: 0.3

3. Set Rotation, Size or Location – Set **Size** X Size = 0.15, Y Size = 0.15, Z Size = 0.3



# 4. Create Flow from Arrival to Pack




1. Create another 2 Queue objects – '**QFailQC**' and '**QReclaim**'.

2. Connect the processor **'QC' to QFailQC using 'A' connect**.

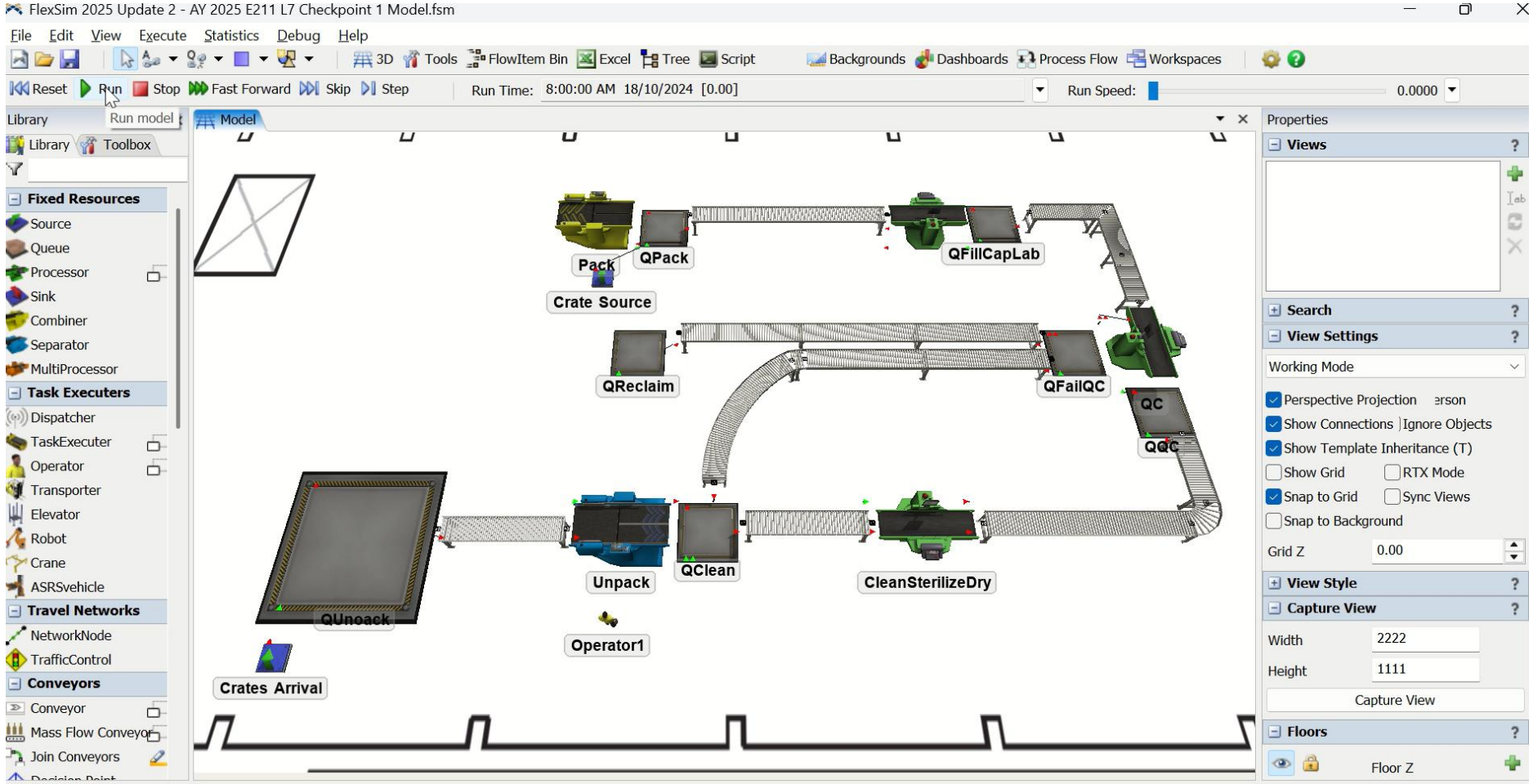
3. Add a conveyor between
   a. QFailQC and QReclaim
   b. QFailQC and QClean
   c. CleanSterilizeDry and QQC
   d. QC and QFillCapLab

4. Connect **the above conveyors to the respective objects (using 'A' connect)** in the direction of flow.

# Model Checkpoint #1

Run the simulation. Do you see the flow on the conveyors from Arrival to Q/Pack and QReclaim?

Your model should look like the simulation below.



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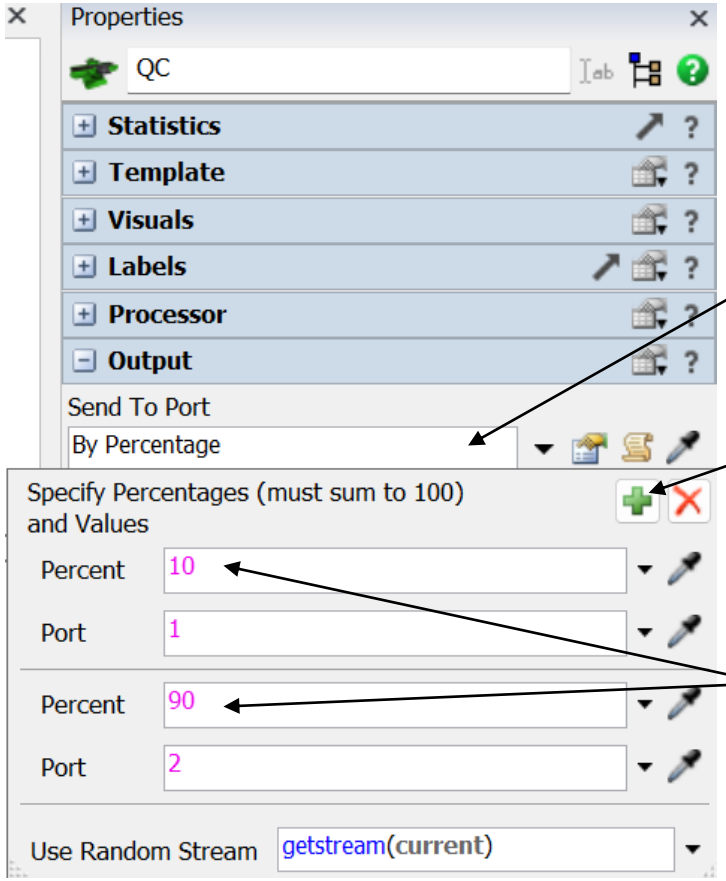
# 5. Set Processor Properties

| | CleanSterilizeDry | FillCapLab | QC |
|---|---|---|---|
| **Max Content** | 12 | 12 | 12 |
| **Process Time** | 10 seconds (constant) | | |

Verify the values of **Max Content** and **Process Time** in **CleanSterilizeDry**, **FillCapLab** and **QC** Processors

# 5. Set QC Properties: Pass/Fail Flow

- When a bottle fails QC (10%), it flows to QFailQC.
- When a bottle passes QC (90%), it flows via conveyor to QFillCapLab.

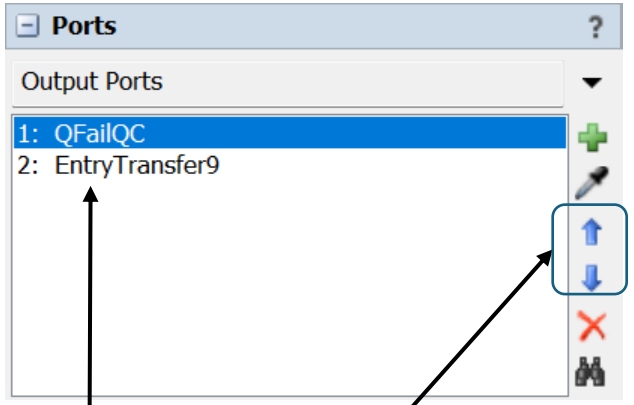



1. Select '**Random**' and then '**By Percentage**'

2. Add Case

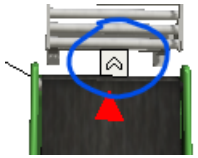
3. Key in the correct percent and port number for each case

4. Check that the output port connections are correct:
- **10% to QFailQC**
- **90% to EntryTransfer (conveyor)**



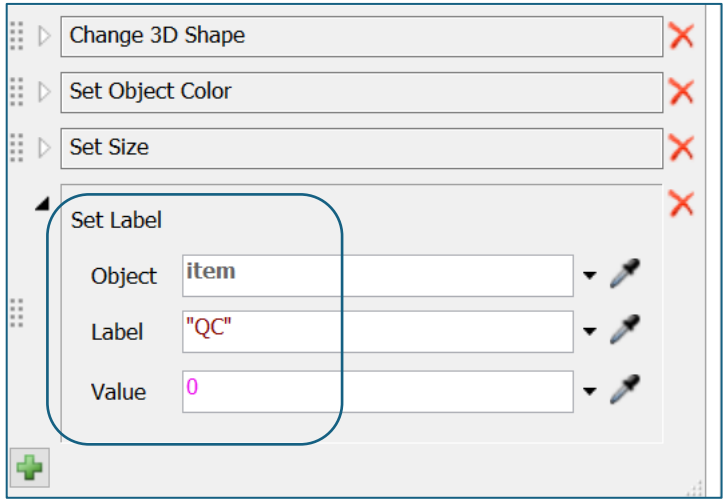

Swap the port numbers if necessary

EntryTransfer ports are found on Conveyor entry points

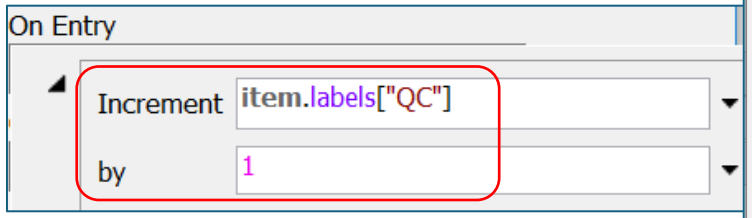


# 6. Track QC Status by Item Label

1. Select **Unpack**, On Exit Trigger → Data → Add Set Label 'QC'

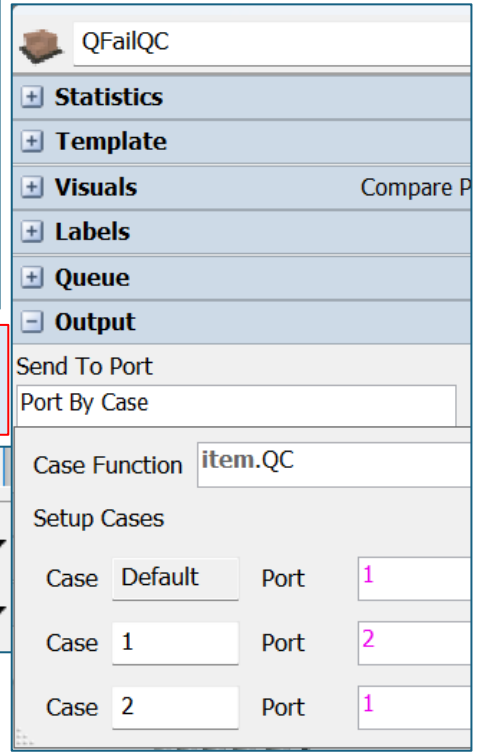


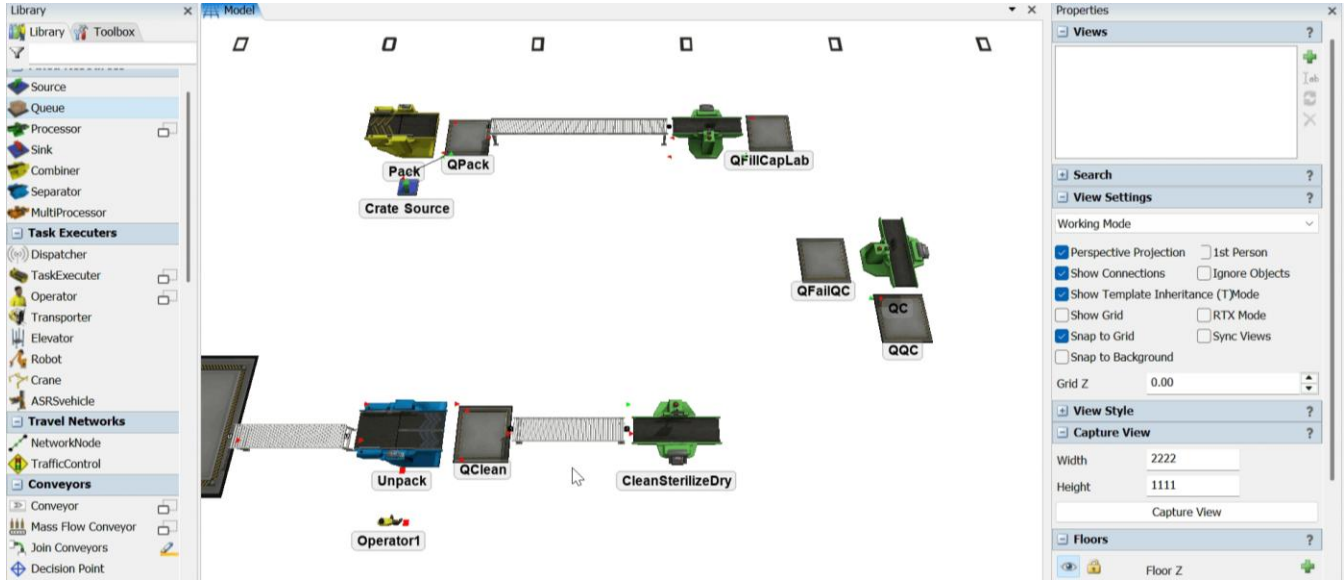

2. Select **QC**, On Entry Trigger → Data → Increment Value of 'QC' label by 1




3. Select **QFailQC**, under Send to Port, select 'Port By Case'. Enter:

- Case Function = **item.QC**
- Setup Cases
  - Case 1: Port 2
  - Case 2: Port 1



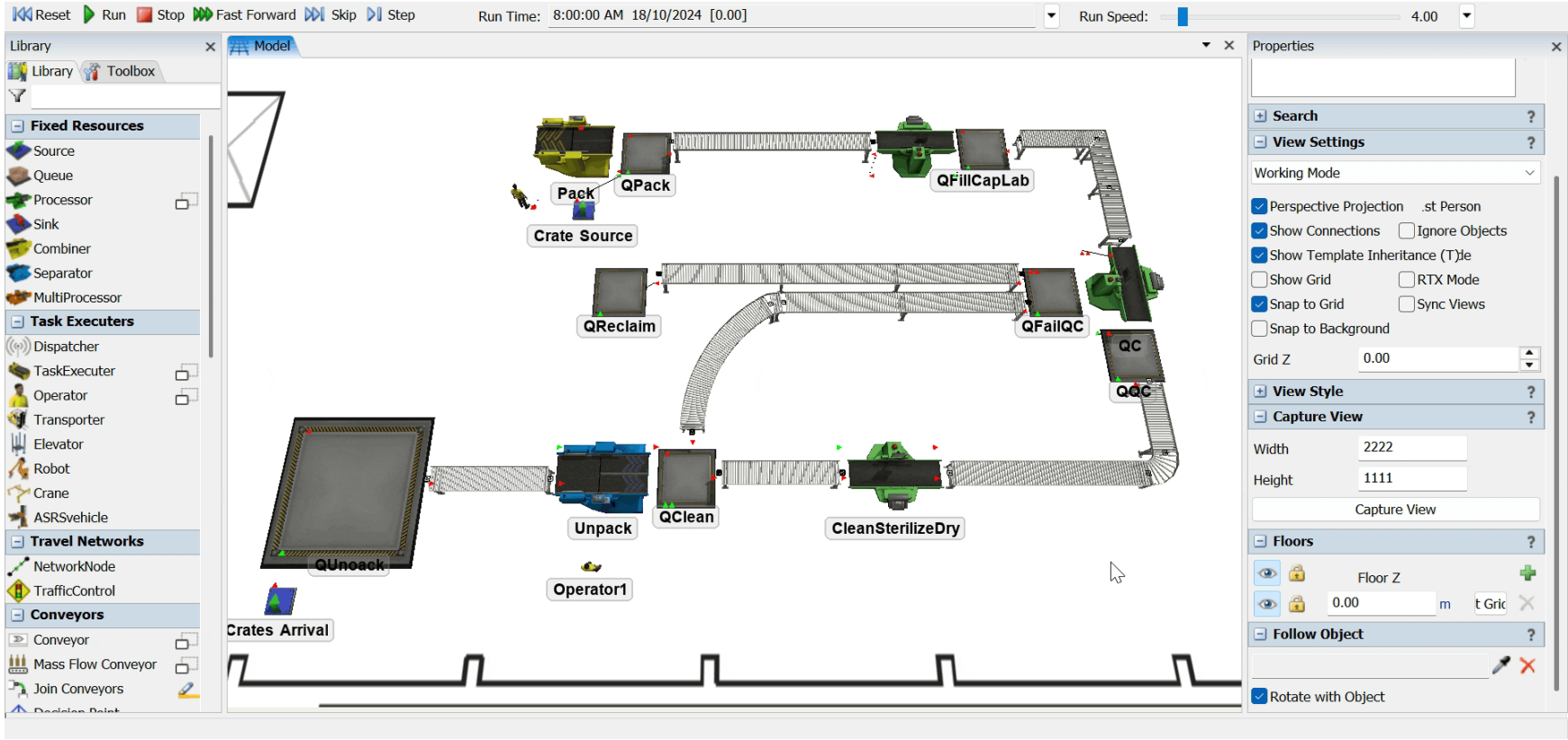



# 7. Set Item Color by QC Status

1. At **QC** Trigger, On Exit trigger → Visual → Set Object Color → Select Color.green

2. At **QFailQC** Trigger, On Entry trigger → Visual → Set Color by Case Value = **item.QC**
1: Color.yellow
2: Color.blue

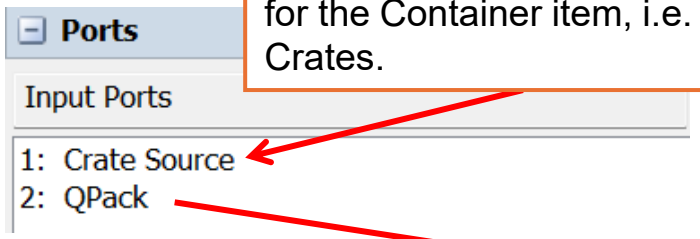
3. At **FillCapLab** processor, Set item color to Silver On Exit.

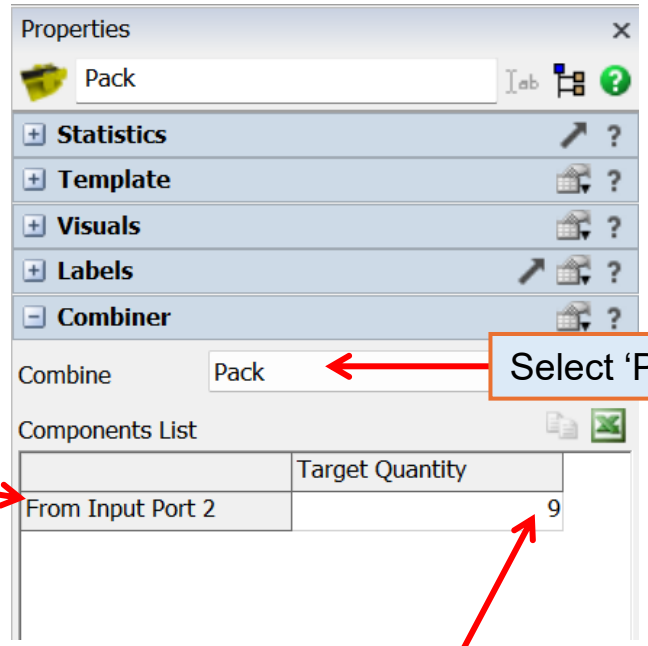


# 8. Combiner to Pack Bottles into Crates

A combiner is used to group multiple items together. It can join items permanently or pack them together so that they can be separated at a later part of the process.

Input port 1 is always reserved for the Container item, i.e. Crates.

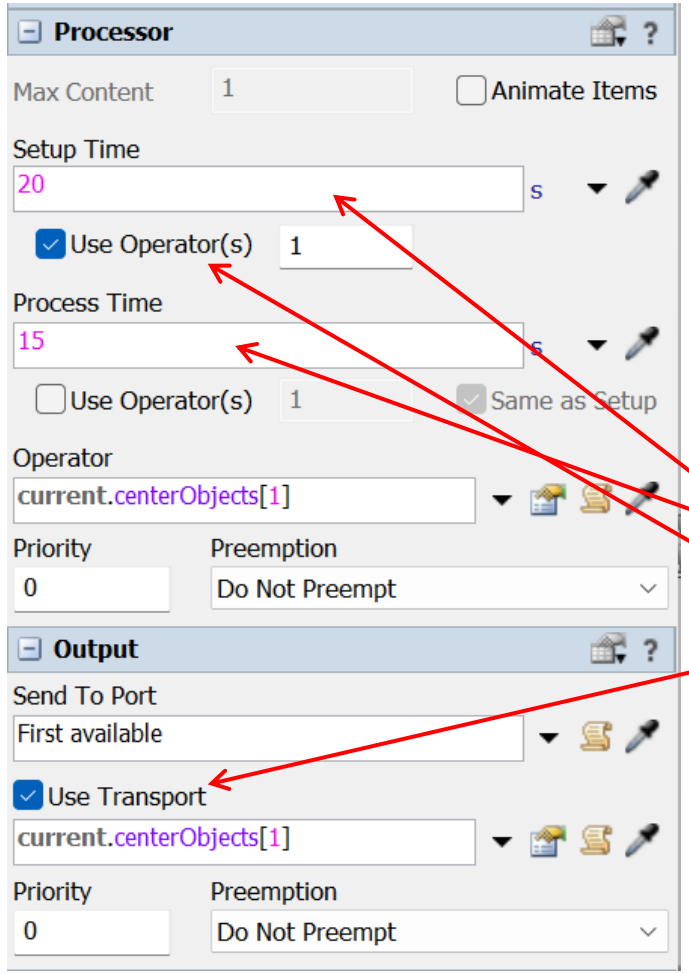



Select ‘Pack’

Key in ‘**9**’ bottles to pack into 1 crate

# Pack Setup and Process Time

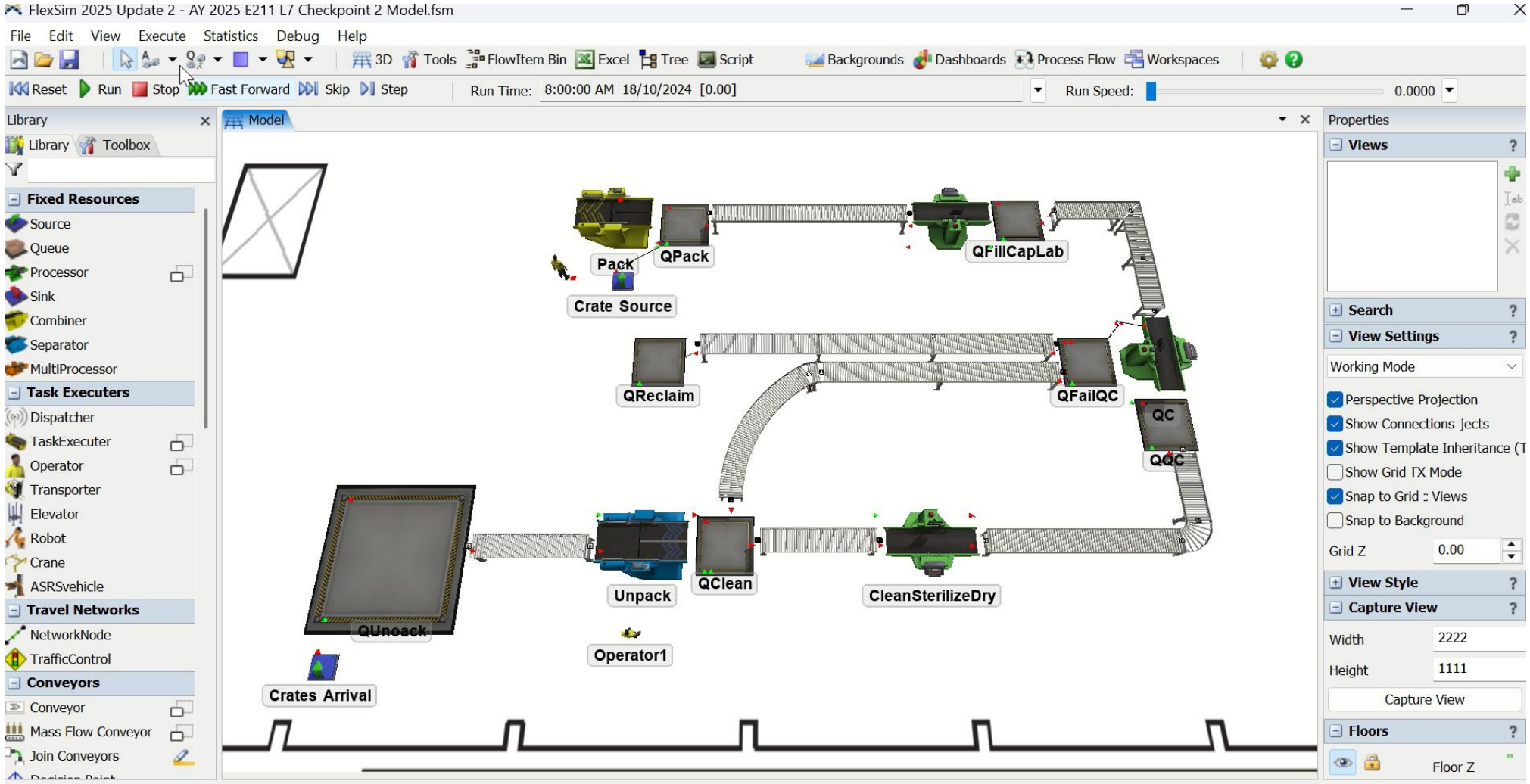



1. Create a new Operator
2. Connect the Operator to Pack via center port connection

3. Key in the Setup and Process timings
4. Check Use Operator for Setup
5. Check Use Transport under Send to Port

# Model Checkpoint #2

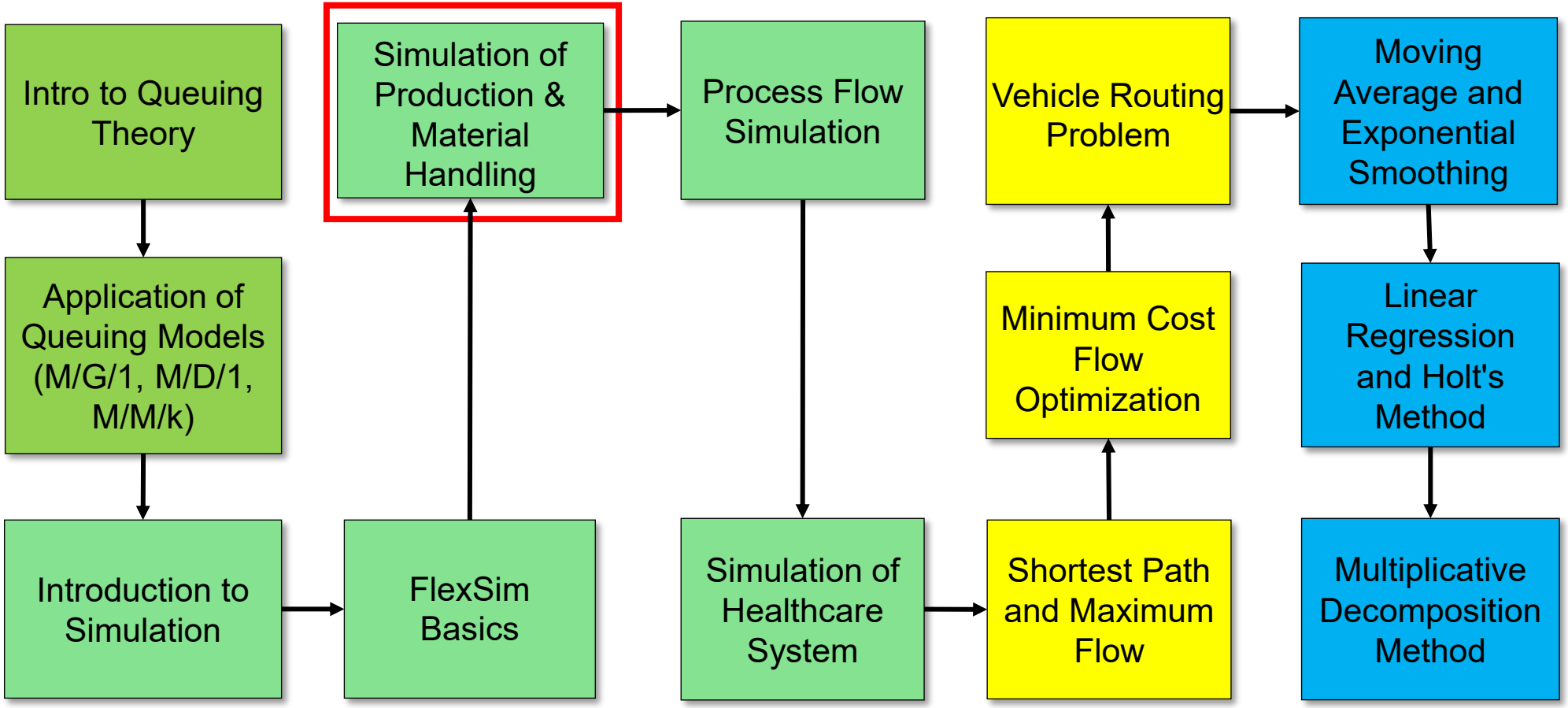
Run the simulation. Observe the flow and transformation of the items into different colours and shapes.



# Quiz

Click the **Quiz** button to edit this object

Deliverable:
Build the 3D bottling model up to checkpoint #2 for this e-Learning lesson. The simulation should run properly.

| | Did not attempt model building | Model is incomplete (unsure how to build certain parts) | Model has errors (bugs) and can't run | Model is completed but simulation looks different from slides | Model is completed and looks correct |
|---|---|---|---|---|---|
| Your response | ○ | ○ | ○ | ○ | ○ |

# E211 Progression